



FUNDAMENTALS OF INFORMATION SECURITY

SESSION 2

COURSE CODE:

CYBE 6123

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Emerging Technologies And How We Live

Learning Objectives



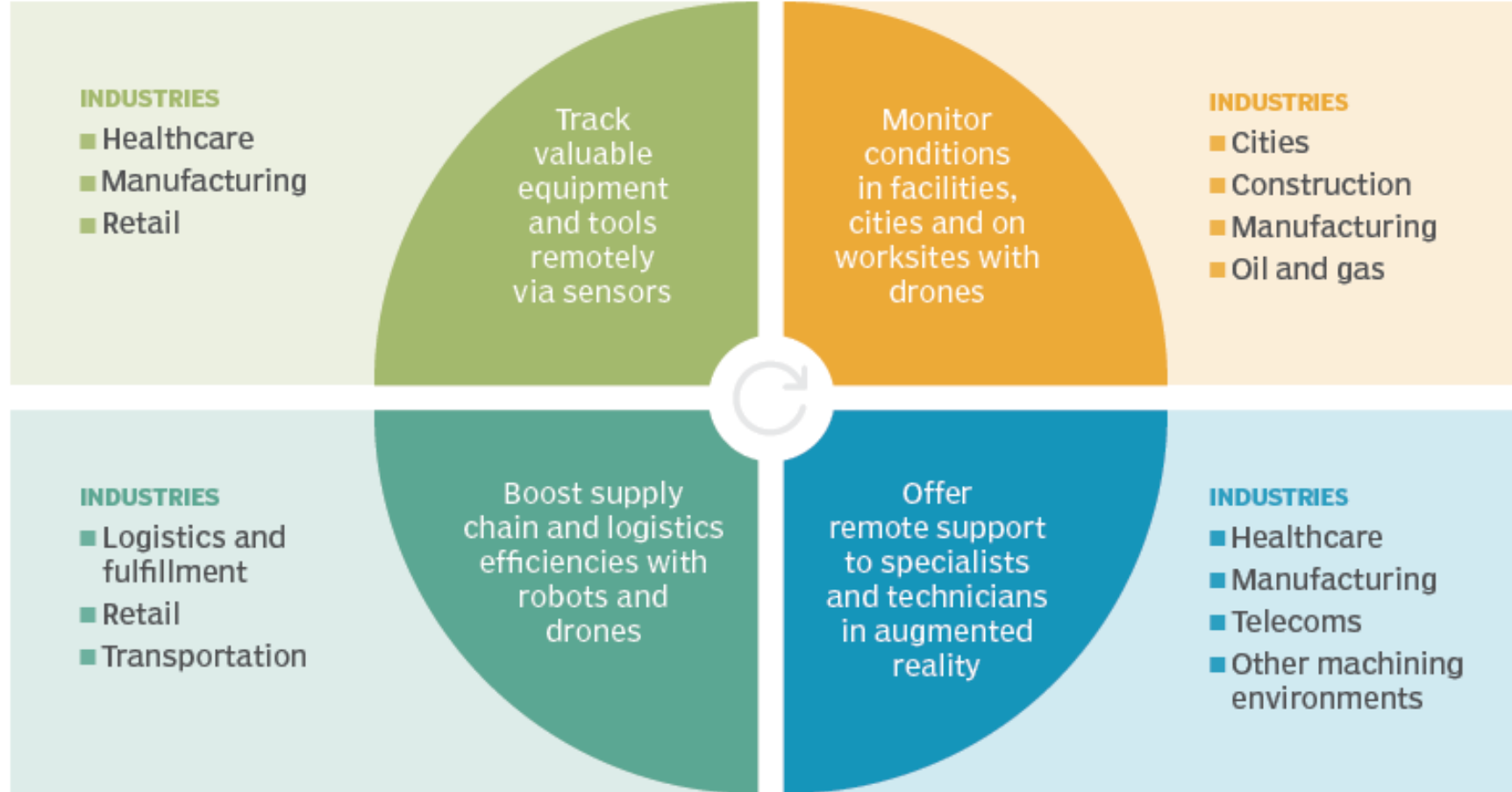
- How the IoT has evolved from the late 1990s to the present
- How the Internet transformed personal and business communications in a TCP/IP (Transmission Control Protocol/Internet Protocol) world
- What the effects of IoT will be on people and businesses and the way we live
- How businesses evolved from brick and mortar to e-commerce to IoT
- Why today's businesses need an Internet and IoT marketing strategy How IP mobility is helping to drive an IoT world
- How mobile applications impact businesses and consumers
- What the key issues created by the IoT are

Emerging Technologies and IoT



- Connected cars.
- Smart appliances.
- Connected security systems.
- Smart agriculture equipment.
- Connected retail.
- Connected healthcare monitors.
- Connected manufacturing equipment.
- Connected cities.
- Connected buildings and building security
- Traffic management
- Smart grids, including smart meters
- Digital payments
- Environmental monitoring
- Supply chain management

Emerging Technologies and IoT



Emerging Technologies and IoT



The Internet today provides a medium for all types of personal and business communications, whether they be:

- **Voice over Internet Protocol (VoIP)**—Real-time voice communications between people, replacing legacy telephone service
- **Instant messaging (IM) chat**—Real-time chat messaging
- **Audio conferencing**—Real-time audio conference calling among multiple people
- **Video conferencing**—Real-time virtual meetings among multiple people
- **Collaboration**—Real-time document sharing and editing with audio-conference and video-conference calling among multiple remote people
- **Digital media**—Audio recordings, pictures, videos for social media up- loading and sharing, or streaming on demand

5 Challenges of IoT



- **Security**—The Internet is already the Wild West, with plenty of bad actors and little law enforcement, yet there is an increasing demand to connect more things to it.
- **Privacy**—How can you control your personal data? Whose data is it? Who owns the intellectual property of personal information, data, and media? What is a privacy policy statement, and why is it important to you?
- **Interoperability**—How do we define standards and protocols such that all IoT-connected devices can communicate and be accessible?
- **Legal and regulatory compliance**—The IoT vision presents legal and regulatory compliance issues that typically have not kept pace with the speed of IoT implementations.
- **Emerging social and economic issues**—The countries of the world and their citizens must quickly learn to understand and overcome the political, environmental, and economic issues presented by the IoT vision.

IoT Impact on Humans



- **Health monitoring and updating**—Sensors can monitor human vital statistics and securely send data analytics to a bedside or mobile application.
- **Home security and smart home control systems**—Homeowners and renters can have real-time access to home security systems
- **Online calendars**—Families and friends can use tools to help plan shared rides and other activities that require coordination.
- **Near real-time tracking and monitoring of friends and family** members via global positioning systems (GPS)

IoT Impact on Humans



- **Online banking, bill paying, and financial transactions**—Anyone can now automate many bill payments using online banking systems and applications via autopay deductions directly from their checking accounts.
- **Online e-commerce purchases for household goods, food, and services**—Consumers have the ability to purchase goods and services on- line and pick them up at a retail store or have them drop-shipped to their front door
- **Automobiles with smart computers and always-on Wi-Fi Internet access**—Car shopping must-haves have evolved beyond miles per gal- lon and now include many advanced features and functions

IoT Impact on Businesses



- **Retail stores**—Stores, banks, restaurants, and manufacturers must have an online presence directly to their customers.
- **Virtual workplace**—Businesses and companies that are in the people or professional services business line do not need to come to a physical office unless it is for important meetings or presentations.
- **Remote sensors** for utility, environmental, and infrastructure monitoring and facility automation
- **City and public service traffic-monitoring applications**—Smart cities can monitor and report on real-time traffic conditions and re-direct traffic flow
- **The IoT is transforming the business-to-consumer (B2C) service delivery model**—Companies are transforming the way they deliver customer service via the Internet.
- **New Anything as a Service IoT applications**—As new IoT and mobile applications are being developed, businesses can implement the Anything as a Service (AaaS).

Businesses Challenges of IoT



- Growing the business through the Internet
- Changing an existing conventional business into an e-business
- Building secure and highly available websites and e-commerce portals
- Building a web-enabled customer-service strategy
- Finding new customers with Internet marketing

e-business strategy

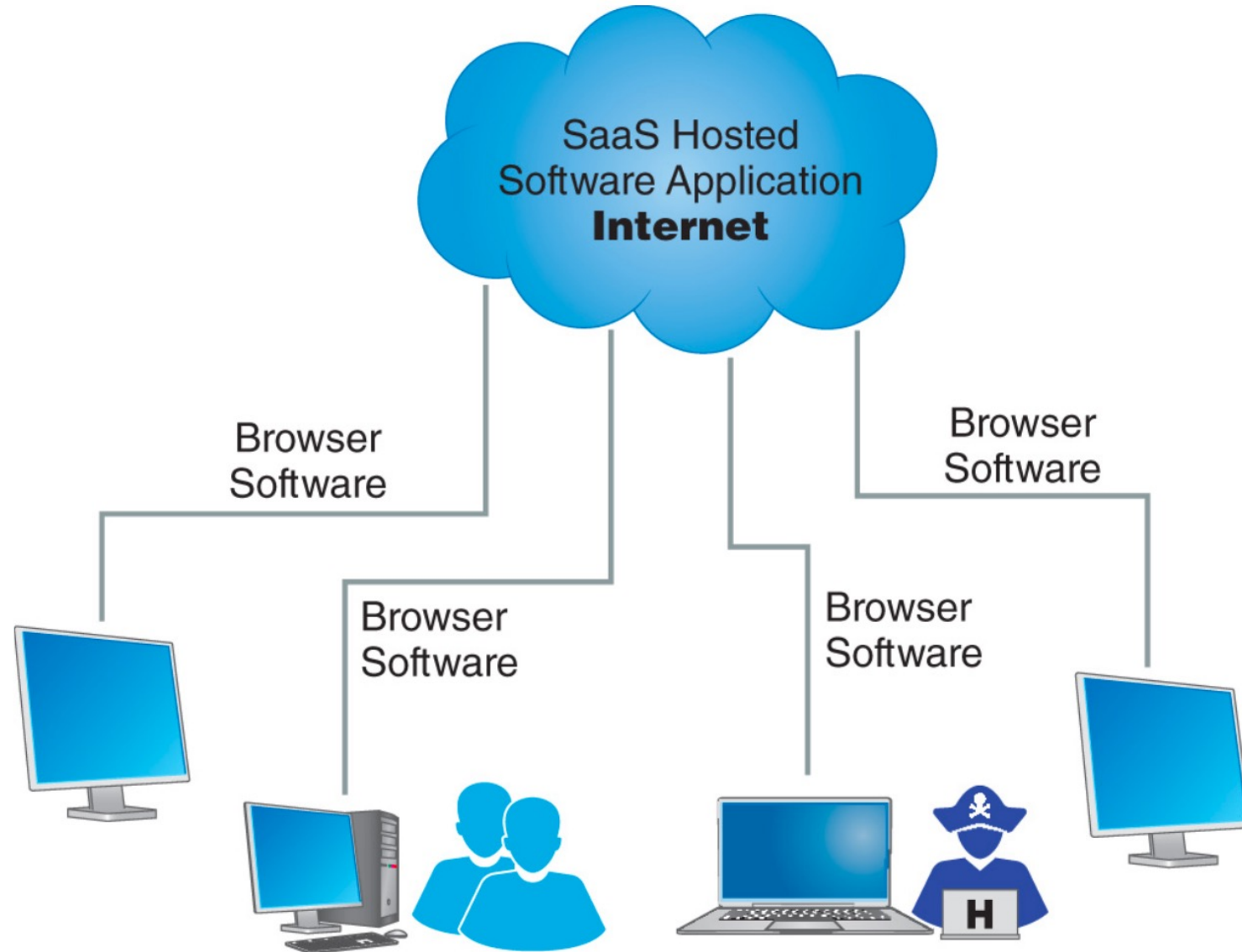


FIGURE 2-1 Software as a Service (SaaS) application delivery model.

e-business strategy



- **E-commerce solution**—This might be an online catalog and system for purchasing goods and services in a secure transaction.
- **Internet marketing strategy**—Internet marketing strategies involve SEO,
- **E-customer service-delivery strategy**—This is a strategy for creating and maintaining a self-serve and online customer service.
- **Payment and credit card transaction processing**—Secure online payment and credit card transaction processing must be encrypted

User and Device Mobility



- **IP Mobility through personal devices**—Devices have changed over time to suit the end user's need for mobility.
- **Need for connectivity** — pushed many companies to accommodate users's push for remaining connected,
- **Mobile applications** – Most applications now allow for mobility right from the planning phase to accommodate end users of all sort
- **BYOD (Bring Your Own Device)**—In response to user's ultimate need to remain connected, BYOD concept has become popular. Hence:
 - Companies allow personal end user devices on enterprise network
 - Put in place necessary controls to safeguard enterprise assets and information
 - Document and train users on BYOD policies to help reduce the risk

Overview of IoT Challenges



- **Security**—How do you keep the bad guys out if you enable the IoT for your personal and professional life?
Privacy—How do you protect your personal identity and private data from theft or unauthorized access, which can lead to identity theft?
- **Interoperability and standards**—How well do IoT manufacturers and ASP developers ensure that devices communicate securely?
- **Legal and regulatory compliance**—What role do the international, federal, and state levels contribute toward legal, tax, and regulatory requirements regarding IoT-related business transactions that involve payment for goods and services?
- **E-commerce and economic development issues**—What are the economic rules of engagement for conducting business on the World Wide Web? How is IoT connectivity and information sharing to be deployed globally?

Security Challenges



IoT devices, such as sensors or consumer items (e.g., Internet-connected TVs and other appliances), are deployed in large quantities.



IoT devices are ubiquitous and can have wide reach into the user or household population.



IoT devices that are not updated or maintained properly may permit vulnerabilities to become entry points into the network or organization



IoT upgrades can be difficult to distribute and deploy, leaving gaps in the remediation of IoT devices or endpoints.

Security Challenges (cont'd)



IoT devices typically do not provide the user or owner with any internal visibility or knowledge of how the device works and connects to the Internet.



IoT devices typically are not physically secure and can be located anywhere in public areas or outside your house.



Environmental IoT devices are typically out in the open capturing readings and measurements for scientific or research reasons.

Privacy Challenges



- **Privacy policy statement**—An actual legal definition of the user's privacy rights as documented in the manufacturer's privacy policy state
- **Definition of data, metadata, or analytical data use and rights**— This is an actual definition that defines what data, metadata, or analytical data is and what it may be used for
- **Ability for a user to provide consent** to a manufacturer's or application service provider's privacy policy statement
- **Determine the domain or extent of privacy**

Privacy Challenges (cont'd)



- How do we address the data source and the data collectors' rights and use of data?
- Is there a happy medium that includes de-identification of private data, with no ability to link that data to a person?
- How will an individual or business even know what a good privacy posture or preference is? Will current social contexts and acceptable behavior extend to the IoT?
- Can we develop privacy by design (with a set of core requirements) and implement a standard of privacy for all of the IoT?

Interoperability and Standards Challenges



- Some manufacturers want to design and deploy proprietary IoT devices and solutions
- Cost factors to implement functional, operational, technical, and security capabilities into IoT devices and applications
- Time-to-market risk
- Technology outdated risk
- A void in interoperability and standards for IoT devices can create an environment of bad IoT devices

Legal and Regulatory Issues



- Compliance with multiple laws and regulations from different countries regarding data sharing and privacy
- Compliance with international standards on technology specifications such as encryption
- Cross-border data transfer and storage
- Devices ensuring proper handling of data according to laws and regulations
- Interconnection with other applications and their compliance status
- IoT device liability issues such as injury, crime accessory etc...

E-Commerce and Economic Development Issues



- **Infrastructure resources**—Foundational to the deployment of the IoT, a communication infrastructure and broadband Internet network are needed within that country.
- **Foundational investments**—Countries seeking to invest in critical infrastructures may be able to leapfrog past other countries
- **Technical and industrial development**—New skills are needed to bring new technologies and economic solutions
- **Policy and regulatory definitions**—Countries and emerging economies are positioned to create and implement policies and regulations in favor of e-commerce

Critical Infrastructure IoT



- Compliance with multiple laws and regulations from different countries regarding data sharing and privacy
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Risks, Threats, and Vulnerabilities

Risk Management and Information Security

- How risk management and information systems security align and what risk terminology is
- What the elements and purpose of risk management are
- approaches for risk assessments and What a risk management plan is
- What a threat is and different types of threats
- What a vulnerability is and how attackers use vulnerabilities to exploit IT assets
- What common attack vectors are and why countermeasures are important

Introduction to Risk Management

- Central focus of information security.
- Key factor in success or failure of most businesses
- There is no risk-free environment
- Risk can be positive (Opportunities) or negative (Loss)
- 2 main principles should always guide your risk management practice:
 - Do not spend more to protect an asset than it is worth.
 - Every countermeasure requires resources to implement and therefore should be aligned with a specific risk.

Risk Terminology

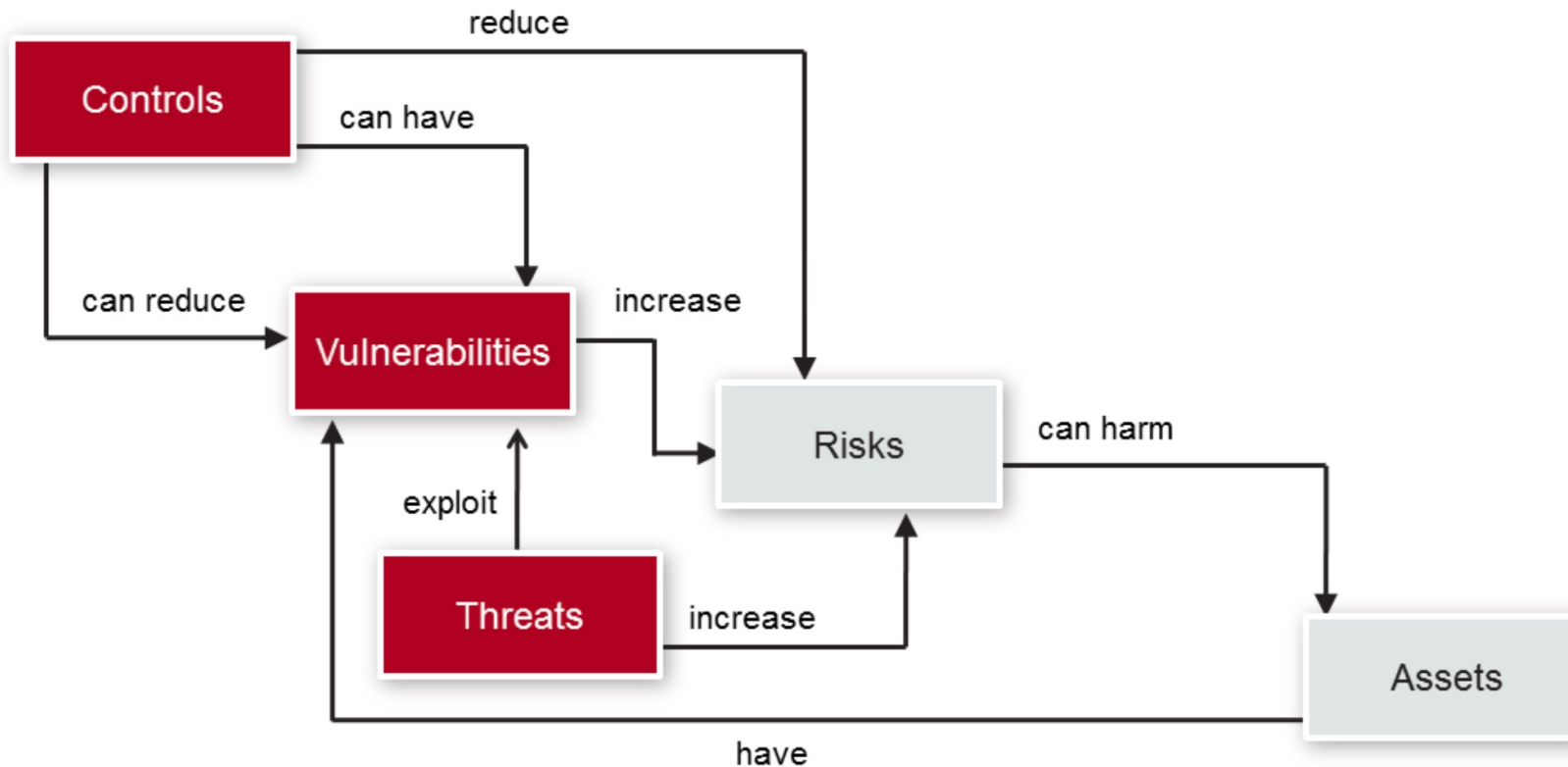
- **Risk**—Risk is the likelihood that something bad will happen that can lead to an eventual change.
- **Threat**—A threat is something bad that might happen to an organization.
- **Vulnerability**—A vulnerability is any exposure that could allow a threat to be realized.
- **Impact**—the amount of risk or harm caused by a threat or vulnerability that is exploited by a perpetrator.

Risk Terminology (cont'd)

- An **event** is a measurable occurrence that has an impact on the business
- An **incident** is any event that either violates or threatens to violate a company's security policy and that justifies a countermeasure;
- **Safeguards or controls** address gaps or weaknesses in the assets or environment that could facilitate the realization of a threat
- **Counter-measures** mitigate or address a specific threat.

Relationships between Security Elements

Overview



Purpose of Risk Management

- The purpose of risk management is to identify possible problems :
 - Before they lead to an incident
 - In time to enable a plan and begin risk-handling activities (controls and countermeasures)
 - On a continuous basis across the life of the product, system, or project
- Establish and maintain risk culture of the organization

Assignment

- Research 10 IoT environments or platforms and provide their importance or how they change the way we live in a short paragraph for each.
- Complete and submit assessment.



Thank You